ResearchOS

A Modular Agent Framework for Integrated Research Workflows

Project Proposal



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1. Background: The Changing Nature of Research Work

Traditional Research

Fragmented and inefficient



Multiple tools, repeated context switching

AI-Agent Supported Research

Integrated and efficient



Unified platform, intelligent collaboration

2. Research Workflow and Pain Points



Key Pain Points

Fragmented tools
Multiple platforms for different tasks

Repetitive operations
Manual setup and repeated execution

Context loss
Research information scattered across tools

Steep learning cost
High barrier to new workflows and domains

3. Existing Systems and Remaining Gap

Project	Strength	Limitation
PaperQA2	Powerful literature understanding	Focuses on paper analysis only
STORM	Automated literature survey	Limited support for experiments and code
AI-Scientist	End-to-end research idea generation	Still experimental and not broadly applicable
Agent Laboratory	Multi-agent collaboration	Lacks unified project management
MLflow / Trackio	Experiment tracking and management	Focuses on experiment tracking, limited research workflow integration



Existing tools are strong in single capabilities, but an integrated project-centric research workspace is still missing.

4. Project Goal and Positioning

ResearchOS

A project-centric agent workspace for research

Unified project context

Bring together papers, code, experiments and insights in one place.

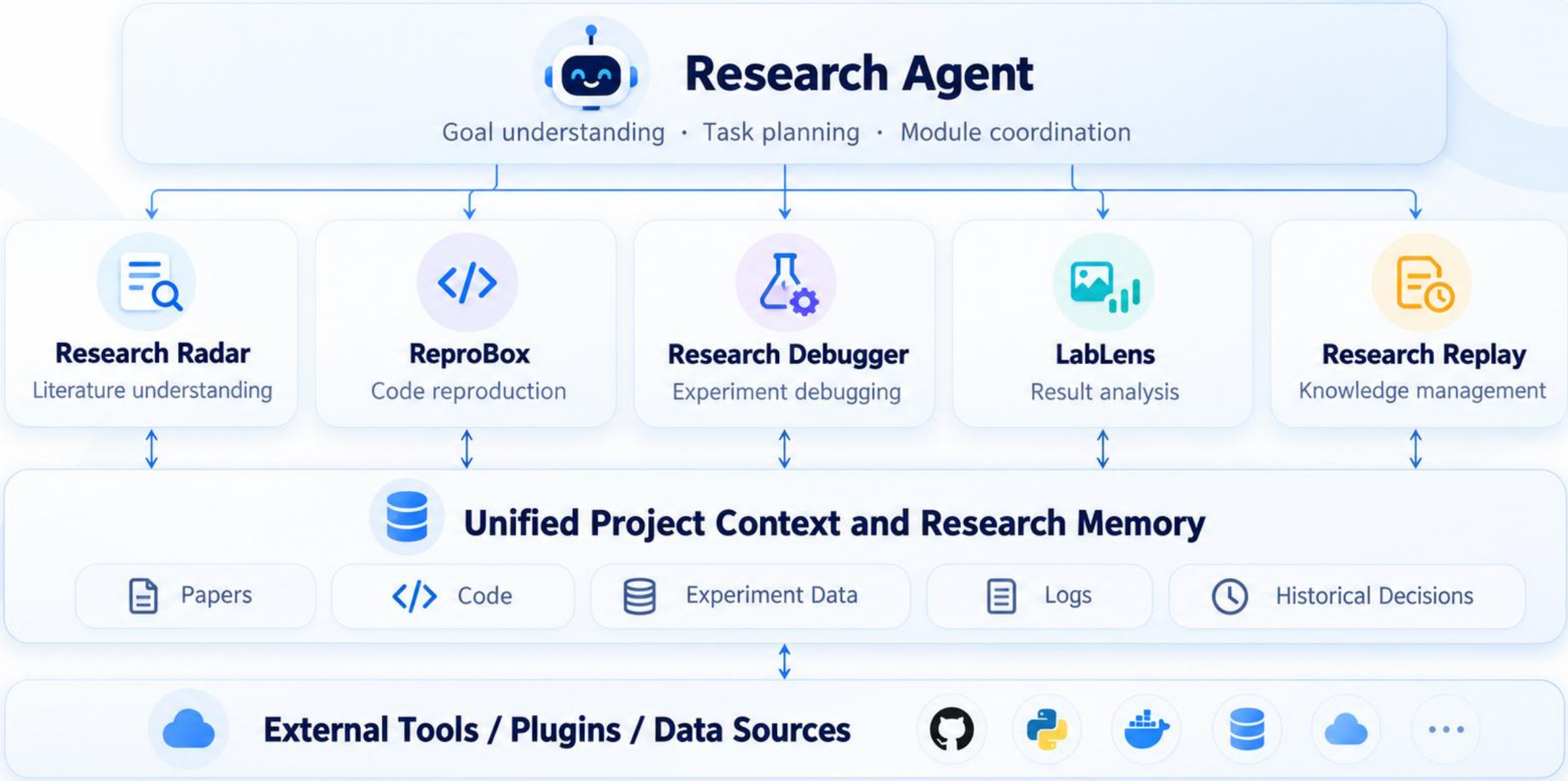
Intelligent task collaboration

Multi-agent collaboration across the research workflow.

Extensible research skills

Easily integrate new tools, models, and domain-specific capabilities.

5. Overall System Architecture



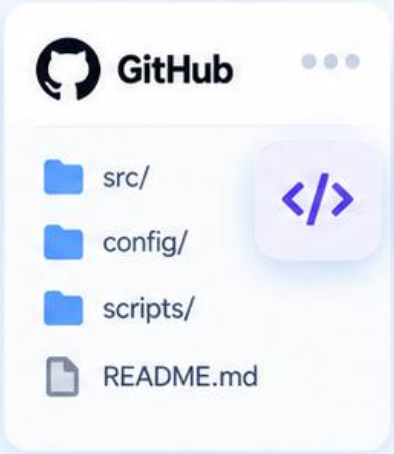
6. Typical Scenario: From Paper to Experiment

1 Discover paper



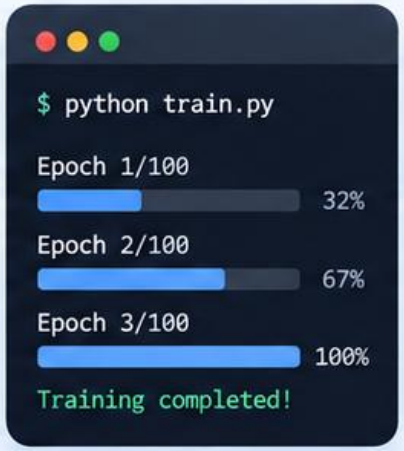
Find relevant papers

2 Reproduce code



Set up and reproduce

3 Run experiment



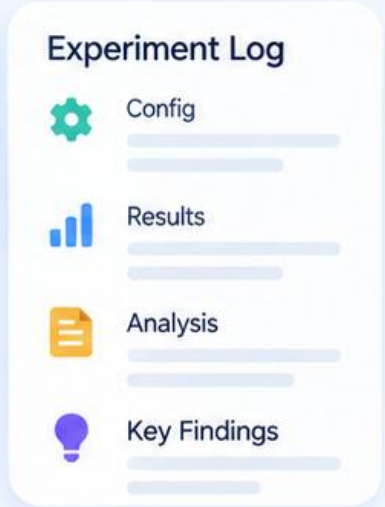
Run experiments and track results

4 Analyze results



Analyze results and generate insights

5 Archive knowledge



Save experiment results and notes

7. Project Highlights



Cross-module context sharing

A unified project context enables seamless information flow across all modules.



Closed-loop research workflow

Covers the full cycle from literature to experiments, analysis, and knowledge management.



Extensible modular framework

Easily integrate new tools, models, and domain-specific research skills.

8. Key Questions and Answers

Q1 **Why not just use ChatGPT, Claude, or Codex?**

A1 They are powerful general tools, but they do not maintain a unified project context across papers, code, experiments, and historical decisions.

Q2 **What is innovative compared with existing research agents?**

A2 ResearchOS focuses on project-centric integration, cross-module collaboration, and persistent research memory rather than isolated single-task assistance.

Q3 **Will five functional modules be too large?**

A3 The system is modular. Development can begin from the core workspace and key modules, then expand step by step.

Q4 **How can we prove the project is useful?**

A4 By validating real use cases such as literature-to-experiment workflows, experiment debugging efficiency, and research history traceability.

9. Roadmap and Expected Outcomes

Stage 1
(1–2 months)

Core workspace and project memory

Stage 2
(3–4 months)

Research task collaboration

Stage 3
(5–6 months)

Multimodal expansion and optimization

Expected Outcomes

- Open-source framework
- Working prototype
- Typical use-case demos
- Extensible skill ecosystem